Learning Trait-Conditioned Expert Decision Trajectories for LLM Agents under Wicked Problems

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Abstract

Expert decision-making under wicked problems requires navigating ambiguous objectives, conflicting values, and evolving constraints through individualized reasoning processes. Although Large Language Models (LLMs) enable expert simulation agents, existing approaches mainly rely on persona descriptions or demonstrations and fail to capture the decision trajectories underlying individual expert behaviors. In this work, we formulate expert simulation as a personalized sequential behavior modeling problem. Using professional design as a representative domain of wicked problem solving, we collect behavioral data from 162 designers and construct a benchmark containing 30-dimensional design traits and reasoning trajectories. We propose a Trait-Conditioned C–K Trajectory Modeling (TC-CKTM) framework, which learns how latent expert characteristics influence dynamic reasoning state transitions and guides LLM agents toward personalized decision behaviors. Experiments across multiple LLM backbones demonstrate that our framework consistently improves reasoning trajectory fidelity, behavioral diversity, and individual-level alignment over existing prompting and fine-tuning based approaches. This work provides a data-driven approach for modeling expert decision behaviors and building personalized intelligent agents for complex real-world scenarios.

CCS Concepts

• **Information systems**;

ACM Reference Format:

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1 Introduction

Many real-world decision-making scenarios are characterized by "wicked problems", where objectives are ambiguous, constraints are dynamically changing, and multiple conflicting values must be simultaneously considered [3] [20]. Unlike well-defined optimization problems, wicked problems do not have a single objectively optimal solution; instead, experts gradually construct decisions through iterative problem framing, knowledge exploration, value negotiation, and compromise formation. Such decision processes are widely observed in domains including design, architecture, urban planning, engineering, and public policy, where expert judgment plays a central role in navigating complex uncertainties [5, 21].

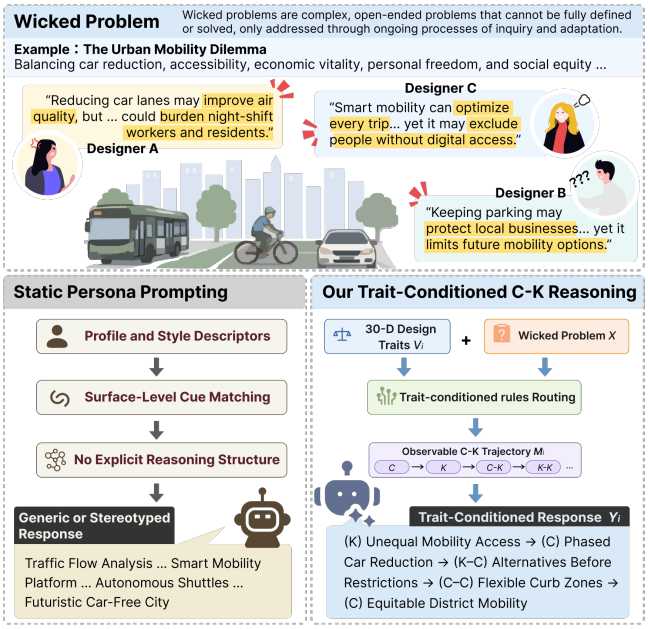


Figure 1: Illustration of the shift from surface-level persona alignment to trait-conditioned C–K reasoning under a wicked problem in design.

Understanding and modeling expert decision behavior under wicked problems is a fundamental challenge for intelligent systems. Existing computational approaches mainly focus on capturing expert knowledge, rules, or final decisions, while the underlying behavioral trajectories that lead experts from an initial dilemma to a final judgment remain largely unexplored. However, for complex decision scenarios, the reasoning trajectory itself contains essential information: different experts may reach similar outcomes through substantially different paths, reflecting their individual preferences, experiences, and value orientations. Therefore, learning personalized expert decision trajectories represents an important problem in human behavior modeling and intelligent decision support.

Recent advances in Large Language Models (LLMs) have enabled new opportunities for simulating human experts and constructing role-playing agents [24]. These agents have shown potential in applications such as user simulation, decision assistance, and interactive systems. Nevertheless, existing LLM-based expert simulation approaches mainly rely on textual personas, background descriptions, or retrieved demonstrations [7, 25, 29]. Such approaches primarily align the surface characteristics of generated responses but do not model the latent behavioral mechanisms governing how

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experts navigate wicked problems. Consequently, generated agents often produce generic expert-like responses and fail to reproduce the individualized decision trajectories of specific experts [6, 10, 31].

From the knowledge discovery perspective, this limitation can be formulated as a personalized sequential behavior modeling problem: given limited observations of an individual's characteristics and historical behaviors, how can we learn the transition patterns that govern their decision trajectories under wicked problems? Solving this problem requires addressing three challenges. First, expert characteristics are latent and cannot be directly observed from decision outcomes. Second, reasoning trajectories are dynamic and situation-dependent rather than fixed behavioral templates. Third, human behavioral data are usually sparse, requiring models that can generalize individual patterns beyond observed scenarios.

To address these challenges, we establish a behavioral modeling framework for wicked problem solving, using professional design as a representative testbed. Design provides a rich environment for studying expert decision trajectories, as designers continuously balance conflicting objectives involving functionality, aesthetics, users, and social values [5, 21]. To characterize latent expert differences, we mine 143 multilingual biographies and monographs of influential designers, extracting recurring professional positions and constructing a 30-dimensional trait representation through clustering and expert validation. Based on behavioral data from 162 professional designers, we further build a benchmark containing individual trait profiles and reasoning trajectories collected under high-conflict dilemmas. Inspired by Cognitive-Affective Personality System (CAPS) theory [18], we propose a Trait-Conditioned C–K Trajectory Modeling (TC-CKTM) framework that learns how expert characteristics influence sequential decision behaviors through structured reasoning states and trait-conditioned transition patterns.

Different from existing LLM-based role-playing approaches that rely on persona descriptions, demonstrations, or alignment constraints [24, 25], our framework models the relationship among expert traits, problem situations, and reasoning transitions, enabling LLM agents to reproduce individualized decision trajectories rather than generic expert-like responses. Extensive experiments across multiple LLM backbones demonstrate improvements in reasoning trajectory fidelity, behavioral diversity, and individual-level alignment, highlighting the effectiveness of trajectory-level expert behavior modeling for LLM agents under real-world scenarios.

Our contributions are summarized as follows:

- We formulate expert decision trajectory modeling under wicked problems as a personalized sequential behavior learning problem and construct a benchmark from 162 professional designers with trait profiles and reasoning trajectories.
- We propose a TC-CKTM framework that learns how latent expert traits influence reasoning state transitions through structured trajectory representations.
- We demonstrate the effectiveness of trajectory-level expert behavior modeling for LLM agents through extensive experiments, achieving improved trajectory fidelity, behavioral diversity, and individual-level alignment across multiple LLM backbones.

2 Related Work

2.1 Human Decision Behavior Modeling

Modeling expert decision behaviors from human data is an important problem in intelligent systems, especially for complex scenarios where decisions emerge through iterative reasoning rather than direct optimization. Design practice provides a representative example of such wicked problem solving, where experts continuously balance conflicting objectives and rely on problem framing, experiential knowledge, and value judgment [5, 16, 21]. Existing knowledge engineering approaches have attempted to formalize design knowledge using structured representations such as knowledge graphs [11]; however, they mainly capture explicit knowledge and lack the ability to model individualized decision trajectories. Cognitive theories such as the Cognitive-Affective Personality System (CAPS) suggest that individual behaviors are jointly influenced by stable traits and situation-dependent patterns [18]. Inspired by these insights, our work studies expert decision trajectory modeling from behavioral observations and learns how latent traits influence reasoning transitions under wicked problems.

2.2 LLM-based Agents and Expert Simulation

Large Language Models (LLMs) have enabled new approaches for simulating human roles and domain experts. Early role-playing agents demonstrated that LLMs can imitate characters through prompt-based role descriptions [24], while subsequent studies improved consistency through character profiles, behavioral constraints, and alignment strategies [7, 25, 27]. Recent surveys further summarize the development of LLM-based role-playing agents and highlight the challenge of maintaining consistent behaviors beyond surface-level text patterns [6, 33, 34].Recent LLM studies in professional design focus on deploying synthetic personas to simulate users or stakeholders for evaluation [15, 17, 19, 22]. However, modeling the expert's internal, non-linear decision-making under high-conflict dilemmas remains unexplored. Existing methods mainly rely on personas, demonstrations, or external constraints to control agent outputs, but they rarely model the latent behavioral mechanisms underlying individual decision differences. Recent works introduce cognitive structures and neuro-symbolic constraints, including Theory of Mind, self-reflection, and ontology-based reasoning mechanisms, to improve agent consistency and reasoning ability [1, 2, 4, 9, 12, 14, 23, 26, 32]. Different from these approaches, our work learns trait-conditioned decision trajectories from human behavioral data and transfers learned patterns to LLM agents for personalized expert simulation.

3 Modeling Expert Decision Trajectories under Wicked Problems

Expert decision-making under wicked problems cannot be characterized solely by final decisions or explicit knowledge. Instead, experts gradually construct solutions through a sequence of reasoning behaviors, including problem interpretation, knowledge exploration, value negotiation, and concept refinement. Meanwhile, different experts may exhibit distinct decision trajectories even when facing the same situation, due to variations in professional experience, personal preferences, and value orientations. Therefore,

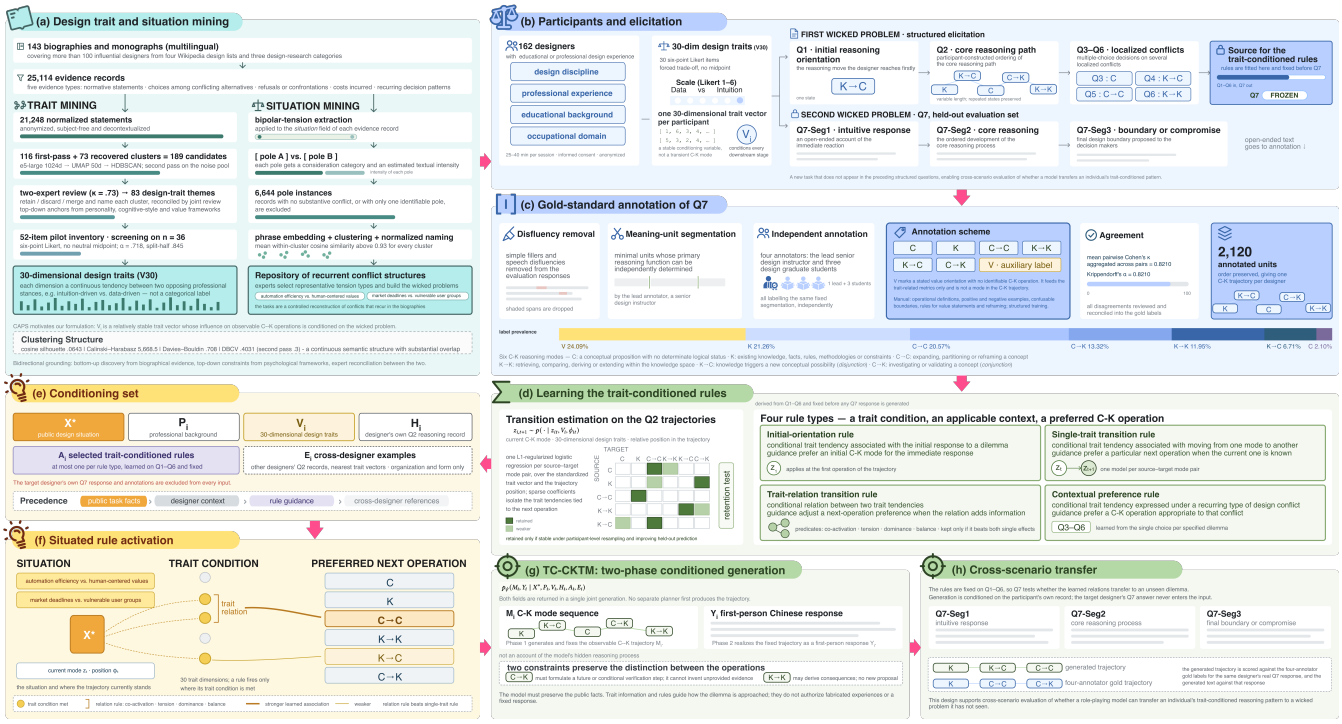


Figure 2: Overview of the proposed framework for learning personalized expert decision trajectories under wicked problems. The framework first extracts latent expert traits from professional knowledge sources, then collects human decision trajectories through wicked problem experiments, and finally learns trait-conditioned reasoning transitions for personalized LLM agents.

this section establishes the representations and behavioral data required for personalized expert trajectory modeling. Specifically, we address three questions: (1) how to represent relatively stable individual characteristics that distinguish different experts; (2) how to represent the dynamic reasoning trajectories through which experts solve wicked problems; and (3) how to collect and annotate behavioral data that capture the relationship between individual traits and decision trajectories.

Figure 2 provides an overview of our framework for learning expert decision trajectories under wicked problems. The framework consists of three stages: (1) extracting latent expert traits and conflict structures from large-scale professional knowledge sources, (2) collecting and annotating expert decision trajectories through controlled wicked problem experiments, and (3) learning trait-conditioned reasoning transitions to enable personalized expert simulation. The central idea is to move beyond modeling final decisions or surface-level personas, and instead learn how individual characteristics shape the sequential behavioral processes underlying expert decisions.

3.1 Problem Formulation

We formulate expert decision trajectory modeling under wicked problems as a personalized sequential behavior learning problem. Given a wicked problem scenario X, an expert i is characterized by a 30-dimensional professional trait V_i and generates a reasoning trajectory M_i together with a final decision Y_i. M_i is represented as a sequence of structured reasoning states:

$$M_i = (m_1, m_2, \ldots, m_T), \quad m_t \in \mathcal{M}, \tag{1}$$

where $\mathcal{M}$ denotes the predefined C-K trajectory space and each m_t represents a specific reasoning operation during decision making. The objective is to learn how expert traits and problem situations influence the transition dynamics among reasoning states. Formally, we aim to model the conditional transition probability:

$$p(m_t | m_{<t}, V_i, X), \tag{2}$$

where $m_{<t}$ denotes previously visited reasoning states. By learning such trait-conditioned transition patterns, the model can generate individualized reasoning trajectories for experts under unseen wicked problems.

Different from conventional decision prediction tasks that focus only on final outcomes Y_i, our goal is to recover the underlying behavioral process that leads to decisions. Therefore, the final first-person decision generation can be formulated as:

$$p(Y_i | M_i, V_i, X). \tag{3}$$

This formulation enables modeling both the intermediate reasoning process and the final decision behavior of individual experts.

3.2 A Bidirectionally Grounded Extraction of Expert Traits

The first challenge in expert trajectory modeling is representing latent individual differences that influence decision behaviors. Inspired by the Cognitive-Affective Personality System (CAPS) theory [18], we model experts through latent traits capturing stable behavioral tendencies with situation dependency. Instead of adopting generic personality categories, we construct domain-specific traits through bidirectional evidence mining and theoretical grounding.

3.2.1 Biographical Data Collection from Influential Designers. We collect biographies and monographs of influential designers across multiple disciplines through an inductive sampling strategy. Representative designers are selected from Wikipedia lists covering industrial design, architecture, graphic design, and urban planning, together with related categories including *Design researchers*, *Design studies*, and *Design history*. This process results in a multilingual corpus of 143 biographies and monographs covering more than 100 designers. Rather than extracting explicit personality descriptions, we identify behavioral evidence reflecting professional positions and decision preferences, including normative statements, conflict-based choices, refusals, professional sacrifices, and recurring decision patterns. This procedure yields 25,114 evidence records containing source information, evidence category, summarized design position, supporting passage, and associated situation.

3.2.2 Semantic Clustering and Trait Discovery. To discover transferable expert tendencies, we anonymize evidence and use a large language model to rewrite each record into a short, subject-free, and decontextualized normative statement. For example, a project-specific description of rejecting ornamentation is normalized as *"Rejects excessive ornamentation and advocates formal simplicity."*. After removing invalid or context-dependent statements, 21,248 normalized statements are retained. Each statement is embedded using `multilingual-e5-large` (1,024 dimensions). UMAP reduces embeddings to 50 dimensions for clustering and two dimensions for visualization. We then apply HDBSCAN with `min_cluster_size=30` and `min_samples=5`. The clustering procedure is shown in Figure 3.

The first clustering stage produces 116 clusters covering recurring professional positions such as accessibility, unity and order, modularity, user experience, and craftsmanship integrity. The clustering quality is evaluated using cosine silhouette score (0.0643), Calinski–Harabasz index (5668.5), Davies–Bouldin index (0.708), and DBCV (0.4031). The positive DBCV score indicates meaningful density structures, while the low silhouette score suggests that design traits form an overlapping continuous space rather than isolated categories. Because 13,006 statements (61.21%) are classified as noise, we perform a second HDBSCAN clustering on the noise pool (DBCV: 0.3000), recovering 73 additional low-frequency but coherent themes. Together, the two stages produce 189 candidate themes for expert review.

3.2.3 Expert Review and Alignment with Psychological Theory. The discovered clusters provide candidate themes but cannot directly determine whether they represent stable and measurable traits. Therefore, two domain experts independently review the 189 themes based on keywords, representative statements, and supporting evidence. Each theme is retained, discarded, or merged and assigned

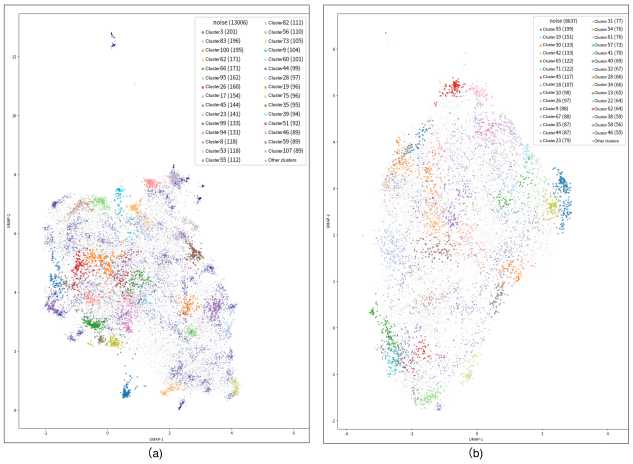


Figure 3: Two-stage density clustering for discovering expert design traits from biographical evidence. The first stage identifies major semantic clusters, while the second stage recovers low-frequency but coherent themes from the noise pool. Points represent UMAP projections, with colors indicating cluster assignments and gray points denoting noise.

an interpretable construct name. The expert agreement reaches Cohen's $\kappa = 0.73$. After resolving disagreements and refinement, we obtain 83 design-trait themes.

To improve construct validity, we introduce established frameworks from personality psychology, cognitive-style research, and value measurement as top-down anchors. These frameworks guide construct boundaries, identify missing dimensions, and improve measurement design without directly equating design traits with generic psychological scales. As illustrated in Figure 2, the trait construction process integrates three sources: bottom-up discovery from biographical evidence, top-down theoretical constraints, and expert reconciliation.

For behavioral experiments, we convert the final dimensions into a measurable inventory. We first develop 52 six-point Likert items, each formulated as a professional trade-off. A pilot study with 36 designers evaluates item quality, achieving Cronbach's $\alpha = 0.718$ and split-half reliability of 0.845. After removing redundant items and refining ambiguous dimensions, we obtain the final 30-dimensional trait inventory. Each dimension represents a continuous behavioral tendency influencing expert decision trajectories rather than a categorical personality type. Detailed definitions and items are provided in the appendix.

3.3 C–K Theory-Grounded Reasoning Trajectory Representation

The 30-dimensional trait profile characterizes *who is making decisions*, however, personalized expert modeling also requires representing *how decisions evolve*. In particular, wicked problems cannot be solved through direct retrieval of existing solutions because experts need to iteratively redefine problems, generate new concepts, acquire knowledge, and evaluate competing constraints. To capture such dynamic reasoning processes, we adopt C–K theory as a

Table 1: Definitions of the six C–K reasoning modes.

Mode	Definition
C	The current reasoning focuses on a conceptual proposition that has not yet acquired a determinate logical status in the existing knowledge space.
K	The current reasoning focuses on existing knowledge, facts, rules, experiential knowledge, methodologies, workflows, or constraints.
$C \to C$	The designer expands, partitions, or reframes a concept by adding, removing, or reorganizing its properties.
$K \to K$	The designer retrieves, compares, derives, or extends information within the knowledge space.
$K \to C$	Existing knowledge triggers a new conceptual possibility, corresponding to a *disjunction* in C–K theory.
$C \to K$	The designer investigates or validates a concept through evidence collection, experimentation, prototyping, or expert consultation, allowing it to gradually acquire a logical status and thereby corresponding to a *conjunction* in C–K theory.

structured representation of expert decision trajectories. C–K theory [8, 13] distinguishes two interacting spaces: the Concept space (C) and the Knowledge space (K). Unlike conventional problem-solving processes that search within an existing solution space, C–K theory views design reasoning as a co-expansion process where concepts and knowledge continuously influence each other. Based on this theoretical foundation, we operationalize expert reasoning trajectories using six observable C–K reasoning modes:

$$\mathcal{M} = \{C,\ K,\ C \to C,\ K \to K,\ K \to C,\ C \to K\}\,. \tag{4}$$

These modes describe different types of reasoning operations during wicked problem solving. The operational definitions are summarized in Table 1. Figure 4 illustrates an example reasoning trajectory under a retail design wicked problem. The trajectory demonstrates how an expert may move among conceptual exploration, knowledge acquisition, concept refinement, and knowledge validation during decision-making.

3.4 Behavioral Data Collection and Annotation

After constructing latent trait and reasoning trajectory representations, we collect behavioral observations that connect individual characteristics with sequential decision processes. As illustrated in Figure 2, the data collection process combines wicked problem construction, expert elicitation, and trajectory annotation.

3.4.1 Construction of Wicked Problems. Biographical evidence is suitable for identifying stable professional positions, but it provides limited access to the fine-grained reasoning processes occurring during decision-making. Therefore, we construct controlled wicked problems to elicit expert reasoning trajectories.

To ground these scenarios in authentic design conflicts, we analyze the *situation* fields extracted from biographical evidence. A large language model is used to identify whether each record contains an abstractable bipolar tension. For each valid tension, the model identifies the consideration category represented by each pole and estimates the strength of textual evidence. Records without

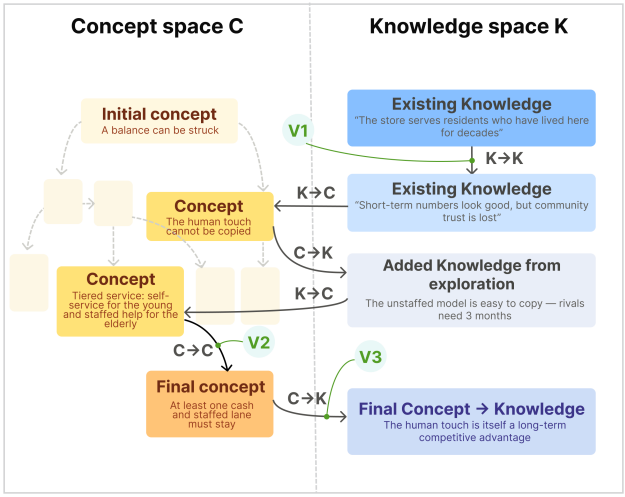


Figure 4: Illustrative C–K reasoning trajectory under a retail design wicked problem. The arrows denote expansions within and transitions between the concept space C and knowledge space K. The green $V1$–$V3$ markers indicate situation-activated trait orientations that condition, rather than constitute, C–K reasoning modes.

substantive conflicts or without identifiable opposing considerations are removed. This process yields 6,644 conflict-pole instances. The extracted conflict expressions are encoded using semantic embeddings and clustered according to semantic similarity. The resulting clusters are manually reviewed and assigned normalized conflict categories.

The mean within-cluster cosine similarity exceeds 0.93 for all retained clusters, indicating strong semantic consistency. The resulting conflict repository provides the basis for constructing representative wicked problems. Domain experts select representative conflict structures and transform them into experimental scenarios involving competing objectives, such as automation efficiency versus human-centered values and market deadlines versus responsibility toward vulnerable user groups.

3.4.2 Participants and Experimental Procedure. We recruited 162 designers with formal design education or professional design experience through online and offline channels. The participants covered multiple design disciplines, including industrial design, architecture, graphic design, and related fields. Their backgrounds varied in professional experience, educational training, and occupational practice, providing diverse individual trait profiles and reasoning behaviors. Each participant completed the experiment under informed consent, and all collected data were anonymized before analysis. Each session lasted approximately 25–40 minutes. The experiment consisted of two major components: trait measurement and wicked problem reasoning elicitation.

First, we collected professional information of participants (PI), then let them completed the 30-dimensional trait inventory introduced in Section 3.2. Each dimension was measured using the refined Likert-scale items obtained from the pilot study. These responses were converted into continuous trait vectors $V_i = [v_1, v_2, \ldots, v_{30}]$

(where each dimension represents an individual behavioral tendency relevant to expert decision-making).

Second, participants solved wicked problem scenarios designed to elicit their reasoning trajectories. The elicitation procedure was designed to capture not only final decisions but also the intermediate reasoning process through which experts navigate conflicting objectives.

The first wicked problem scenario followed a structured elicitation procedure consisting of several reasoning-oriented questions: **Q1:** Initial reasoning orientation selection, where participants identified their primary consideration when approaching the problem. **Q2:** Core reasoning path construction, where participants organized the major reasoning steps underlying their decision process. **Q3–Q6:** Localized conflict decisions, where participants responded to multiple sub-scenarios involving different value trade-offs.

These structured responses were used to analyze general patterns of expert reasoning trajectories and the relationship between trait dimensions and reasoning transitions. To evaluate whether learned expert patterns could transfer beyond observed scenarios, we designed a second wicked problem scenario as a held-out evaluation task:

Q7 introduced a new problem setting that was not included in the training-oriented elicitation process. Participants were required to provide an ordered description of their decision-weighing reasoning process. This design enables cross-scenario evaluation of whether the model can reproduce an individual's trait-conditioned reasoning trajectory when encountering an unseen wicked problem.

3.4.3 Gold-Standard Annotation of the Evaluation Set. To construct reliable trajectory labels, we annotate the held-out Q7 responses at the reasoning-unit level. After removing fillers and speech disfluencies, a senior design instructor segments responses into minimal reasoning units, which are independently labeled by three additional design graduate students following the annotation guidelines.

The annotation scheme follows the six C–K reasoning modes defined in Section 3.3. We additionally introduce an auxiliary label V for explicit value-oriented statements without identifiable C–K operations; V is only used for trait-related evaluation and is not considered a reasoning mode. Annotators receive structured training and detailed guidelines covering definitions, examples, and ambiguous cases. Annotation reliability is evaluated using pairwise Cohen's κ and Krippendorff's α. Both metrics reach 0.8210, indicating substantial agreement. After resolving disagreements, the final evaluation set contains 2,120 annotated reasoning units with the following label distribution: V: 24.09%, K: 21.26%, $C \rightarrow C$: 20.57%, $C \rightarrow K$: 13.32%, $K \rightarrow K$: 11.95%, $K \rightarrow C$: 6.71%, and C: 2.10%. This annotated dataset provides the basis for evaluating whether a model can recover expert-specific reasoning trajectories rather than merely imitate final response styles.

4 Trait-Conditioned C–K Trajectory Modeling

Based on the representations introduced in Section 3, We model individualized designer role-playing as the joint generation of an observable C–K trajectory and a first-person decision with its justification. Given a public design situation X^*, generation is conditioned on designer i's professional background P_i, Q2 reasoning record H_i, situation-adaptive trait cues $\widetilde{V}_i(X^*)$, selected trait-conditioned rules R_i learned from Q1–Q6, and aligned examples E_i:

$$p_\psi(M_i, Y_i \mid X^*, P_i, \widetilde{V}_i(X^*), H_i, R_i, E_i). \tag{5}$$

The complete 30-dimensional profile V_i introduced in Section 3.2 is used internally for trait and rule selection, but is not directly serialized into the prompt. Instead, $\widetilde{V}_i(X^*)$ contains the trait dimensions that are most relevant to the target situation and the target designer. E_i contains three state–text-aligned Q7 examples from other designers; these examples are used to calibrate C–K operation semantics and output granularity rather than provide answers to imitate. The target designer is excluded from E_i, and their Q7 response is never provided. Public task facts always take precedence over all conditioning signals.

Table 2: Trait-conditioned rule families.

Rule type	Condition	Guidance
Single-trait transition	Trait tendency associated with a transition.	Prefer a next operation given the current operation.
Trait-relation transition	Co-activation, tension, dominance, or balance between two traits.	Adjust the next-operation preference beyond separate trait effects.
Contextual preference	Trait tendency under a recurring design conflict.	Prefer an operation appropriate to that conflict.

4.1 Learning Trait-Conditioned Rules

We learn trait-conditioned reasoning rules from Q1–Q6 and freeze them before Q7 generation. The rules express preferences over subsequent operations and situation-specific choices; none specifies a deterministic transition. The Q7 reasoning-response method uses the three families in Table 2.

For a Q2 trajectory, let z_{it} denote designer i's C–K operation at position t. For each source–target pair, we fit an L1-regularized logistic regression over the standardized V30 profile and relative trajectory position:

$$z_{i,t+1} \sim p(\cdot \mid z_{it}, V_i, \phi_{it}). \tag{6}$$

Sparse coefficients identify trait tendencies associated with the next operation after accounting for the current operation and position. Candidate rules are filtered using participant-level support, bootstrap stability, sign consistency, and held-out evidence.

Trait-relation rules are learned in two stages. We first identify stable trait pairs using participant-level rank correlations. For each pair, we construct interpretable predicates—co-activation, tension, dominance, and balance— and test whether the predicate adds predictive information beyond the two individual trait effects. Q3–Q6 provide contextual operation choices and are used to learn contextual preferences rather than artificial transitions. These rules are empirical soft preferences, not causal or universal psychological laws. Detailed thresholds and rule statistics are provided in Appendix B.

4.2 Conditioned Joint Generation

For designer i, the selector uses the complete V_i internally and retains at most one active rule. If single-trait and relation rules conflict for the same source operation, the relation rule receives soft priority. Rules may be ignored when inapplicable, and public facts always override rule guidance.

In particular, $C \rightarrow K$ formulates a future or conditional verification step without inventing evidence, whereas $K \rightarrow K$ derives consequences from known facts without introducing a new proposal. The concatenated Y_i is a coherent first-person response.

5 Experiments

5.1 Evaluation Benchmark

We use behavioral data from 162 designers. Information collected in Questions 1–6 (Q1–Q6) is used to construct each participant's professional background, 30-dimensional value profile (V30), and historical response to a design problem. Question 7 (Q7) presents a previously unseen dilemma in design. The target output is participants' response to Q7.

5.2 Compared Methods

We compare our method with three prompt-only baselines and two trainable role-playing baselines. Prompt-only baselines are evaluated on Qwen3-8B, Llama-3.1-8B-Instruct, Gemini 3 Flash, and GPT-4o. CoSER and RAR-RIA are adapted using Qwen3-8B and Llama-3.1-8B-Instruct.

Prompt-only baselines. All prompt-only baselines generate both a C–K state sequence and a natural-language answer in one response.

SP uses only the Q7 situation and a generic designer role, without participant-specific information.

PP additionally provides the target participant's personal and professional information (PI), including design field, educational background, work context, project role, and design experience.

ICL-Q2 extends PP with three Q2 demonstrations from other participants. Each demonstration contains PI, a Q2 C–K state sequence, and a Q2 natural-language response. The target participant is always excluded from the demonstration set.

Trainable role-playing baselines. We adapt CoSER [30] and RAR RIA [28] to the designer questionnaire data and train them on Q1–Q6 using their respective LoRA supervised fine-tuning procedures. CoSER uses its `model_speech` field as the prediction, whereas RAR-RIA uses its `model_answer` field.

For these two trainable baselines, we evaluate only the generated natural language. We do not compute KC state metrics for CoSER or RAR-RIA, because their original methods do not define a KC mode output.

5.3 Configuration of our Method

Our method performs trait-conditioned aligned C–K generation in a single model call. Tt is evaluated on Qwen3-8B, Llama-3.1-8B-Instruct, GPT-4o, Claude Sonnet 4.6, and Qwen3-235B-A22B-Instruct-2507. GPT-4o is the development and ablation backbone. We use temperature 0.7, top-p 0.9, and at most 1,024 new tokens for generation.

Table 3: Evaluation metrics. All metrics are computed on the Q7 reasoning response.

Category	Metric	Meaning
Response alignment	BGE own-user margin	Similarity to the target gold response minus average similarity to other gold responses. Higher is better.
Trajectory fidelity	KC-NES	Normalized edit similarity between generated and gold C–K sequences. Higher is better.
Trajectory personalization	KC own-user AUC	How often the generated sequence is closer to its own gold sequence than to another participant's gold sequence. Higher is better.
	KC own-user margin	Similarity to the target gold sequence minus average similarity to other gold sequences. Higher is better.
Value-expression faithfulness	Claim P / R / F1	Match between generated V30 claims and human V30 claims for the same participant. Higher is better.
	Profile-polarity precision	Fraction of generated claims whose direction matches the participant's V30 profile. Higher is better.
	Own-profile advantage	Alignment with the target V30 profile minus average alignment with other profiles. Higher is better.
Anti-collapse	KCDR	Fraction of human trajectory diversity retained by generated trajectories. Higher is better.
	TGC	Agreement between generated and gold participant geometry. Higher is better.
	Top share	Frequency of the most common generated trajectory. Lower usually means less collapse.

The prompt contains (i) three other-participant Q7 reasoning-response examples, (ii) target participant's PI, (iii) the participant's 4 most significant traits among the 30 dimensional traits, (iv) the target participant's complete Q2 state sequence and response, (v) the full Q7 situation and reasoning-response instruction prompt.

The model returns a single JSON object contains the KC trajectory and final response.

5.4 Evaluation Metrics

We evaluate 5 properties,which you can see in table **??**.

5.5 Main Results

Table 4 reports the metrics most directly related to personalization and trajectory collapse. Generic prompting frequently converges to a small set of population-level chains. On Qwen3-8B, SP produces only one sequence for all 162 participants, whereas our method produces 50 unique sequences and reduces TopShare from 1.0000 to 0.1173. It also achieves the highest KCDR (0.5472) and BGE own-user margin (0.0035). The same anti-collapse pattern holds on GPT-4o: our method yields 132 unique sequences, lowers TopShare to 0.0309, and attains the highest KCDR (0.5904). These results

Table 4: Complete automatic results on 162 participants. Higher is better for BGE, EM, KC-NES, KC-AUC, KC-Margin, KCDR, TGC, and Unique; lower is better for LenErr and TopShare. "–" indicates that the corresponding method was not run on that backbone or, for CoSER and RAR-RIA, that the method does not produce a C–K state sequence.

Backbone	Method	BGE	EM	LenErr	KC-NES	KC-AUC	KC-Margin	KCDR	TGC	Unique	TopShare
Qwen3-8B	SP	−0.0000	0.0000	1.4444	0.2977	0.5000	0.0000	0.0000	0.0000	1	1.0000
	PP	−0.0000	0.0062	2.0370	0.2773	0.5047	0.0067	0.3953	0.0340	20	0.3889
	ICL	−0.0017	0.0000	3.6173	0.2435	0.4961	−0.0031	0.3811	−0.0134	47	0.2037
	CoSER	−0.0056	–	–	–	–	–	–	–	–	–
	RAR-RIA	−0.0075	–	–	–	–	–	–	–	–	–
	Ours	**0.0035**	0.0000	1.6049	0.2521	0.5011	−0.0064	**0.5472**	−0.0323	**50**	**0.1173**
Llama-3.1-8B	SP	−0.0000	0.0000	4.3210	0.2522	0.5000	0.0000	0.0000	0.0000	1	1.0000
	PP	0.0004	0.0000	7.6852	0.2550	0.5355	0.0129	0.7783	0.0070	154	0.0123
	ICL	−0.0073	0.0000	5.0185	0.2589	0.5153	0.0049	0.7580	−0.0213	156	0.0123
	CoSER	−0.0035	–	–	–	–	–	–	–	–	–
	RAR-RIA	0.0040	–	–	–	–	–	–	–	–	–
	Ours	−0.0013	0.0000	2.4198	0.2202	0.4940	−0.0051	0.7002	0.0523	119	0.1728
GPT-4o	SP	0.0030	0.0000	1.5926	0.2764	0.5194	0.0134	0.3482	0.0207	40	0.4506
	PP	−0.0012	0.0000	1.5679	0.2650	0.5042	0.0031	0.3908	0.0209	43	0.3457
	ICL	0.0003	0.0000	2.3642	0.2504	0.5056	0.0029	0.5888	0.0115	121	0.0494
	Ours	−0.0017	0.0000	3.8272	0.2638	0.4997	−0.0033	**0.5904**	−0.0100	**132**	**0.0309**

Table 5: Complete GPT-4o ablation results on 162 participants.

Configuration	BGE	EM	LenErr	KC-NES	KC-AUC	KC-Margin	KCDR	TGC	Unique	TopShare
Ours	−0.0017	0.0000	3.8272	0.2638	0.4997	−0.0033	0.5904	−0.0100	132	0.0309
Ours w/o rules	−0.0008	0.0000	3.8580	0.2674	0.5066	0.0035	0.5736	0.0137	139	0.0247
Ours + Full30	−0.0007	0.0000	3.8148	0.2632	0.4996	−0.0021	0.5852	−0.0011	138	0.0247
Ours w/o Q2	−0.0009	0.0000	3.1543	0.2706	0.5011	0.0009	0.5732	0.0131	125	0.0617
Ours w/o active4	0.0017	0.0000	3.8765	0.2688	0.5131	0.0032	0.5646	−0.0081	130	0.0494
Ours w/ shuffled rules	0.0018	0.0000	3.6049	0.2674	0.5037	0.0028	0.5974	0.0149	134	0.0370

show that the proposed conditioning scheme substantially reduces convergence to a dominant population-level trajectory.

The BGE result is backbone-dependent, but the trajectory-level evidence is consistent across Qwen3-8B and GPT-4o: our method expands the support of generated C–K trajectories while reducing the frequency of the dominant chain. We therefore interpret the main gain as resistance to population-level mode collapse.

5.6 Ablation Analysis

Table 5 isolates the components most relevant to this effect. Removing Q2 increases TopShare from 0.0309 to 0.0617 and reduces Unique from 132 to 125, showing that participant history helps differentiate trajectories. Removing the learned rules reduces KCDR from 0.5904 to 0.5736, indicating that rule-guided routing preserves more of the between-participant variation found in the human trajectories. Finally, adding Full30 provides no KCDR gain over active4-only (0.5852 vs. 0.5904), supporting the more compact active-trait representation. Overall, Q2 contributes most clearly to avoiding a dominant shared chain, while the rule library improves diversity retention. Full30 does not add a corresponding benefit, so active4-only is retained as the final, more token-efficient prompt configuration.

6 Conclusion

In this paper, we proposed a trait-conditioned expert trajectory modeling framework for simulating personalized decision behaviors under wicked problems. By integrating latent 30-dimensional design traits with explicit C–K reasoning modes, our framework captures how individual characteristics shape sequential reasoning transitions and guides LLM agents to generate expert-consistent decision trajectories. Extensive empirical evaluations across multiple scenarios demonstrate that our framework consistently outperforms standard prompt-based baselines, with the trait-conditioned reasoning transition mechanism playing a critical role in improving trajectory fidelity, preserving behavioral diversity, and achieving individual-level alignment. Furthermore, our framework demonstrates robust generalizability across diverse LLMs with different scales and providers, and its effectiveness remains stable after model adaptation through parameter-efficient fine-tuning. These results indicate that learning expert behaviors from human decision trajectories provides an effective approach for building intelligent agents. Ultimately, this work bridges human behavioral modeling and LLM agents, providing a scalable paradigm for transferring expert decision patterns to intelligent systems in real-world scenarios.

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A Evaluation Metrics and Statistical Protocol

We evaluate the Q7 reasoning response (Seg2) on the fixed cohort of 162 participants. The proposed method returns one ordered array of aligned units, where each unit contains a C–K state and the text realizing that state. The state alphabet is $\mathcal{Z} = \{K, C, K{\to}C, C{\to}K, K{\to}K, C{\to}C\}$. V30 denotes the participant's value profile and is not a trajectory state. Let $M_u = (m_{u1}, \ldots, m_{uT_u})$ and $\widehat{M}_u = (\widehat{m}_{u1}, \ldots, \widehat{m}_{u\widehat{T}_u})$ be the gold and predicted trajectories for participant u, respectively. Auxiliary boundary symbols, if present, are removed before evaluation. Let Y_u and $\widehat{Y}_u$ denote the corresponding gold and generated responses. Metrics are grouped into trajectory fidelity, participant-specific personalization, and anti-collapse diagnostics; high diversity alone is not interpreted as fidelity or personalization.

A.1 Textual Own-User Fidelity

BGE own-user margin. We encode every response with the frozen BGE-M3 dense encoder $e(\cdot)$ (1024 dimensions; L2-normalized). For

$$q_{uv} = \cos\Big(e(\widehat{Y}_u), e(Y_v)\Big),$$

the participant-level margin is

$$b_u = q_{uu} - \frac{1}{N-1}\sum_{v\neq u} q_{uv}, \qquad \text{BGE-Margin} = \frac{1}{N}\sum_u b_u. \quad \text{(A.1)}$$

A positive value means that a generated response is, on average, closer to its intended participant's gold response than to the gold responses of the other participants. This is an identity-specific text metric, rather than a generic semantic-similarity score. We also retain mean own-user and other-user similarities for interpretation.

A.2 C–K Trajectory Fidelity

Exact match and length error. Exact match requires equality of the complete ordered state sequences:

$$\mathrm{EM} = \frac{1}{N}\sum_{u}\mathbb{I}[\widehat{M}_u = M_u]. \tag{A.2}$$

Length error measures only the discrepancy in the number of operations:

$$\mathrm{LenErr} = \frac{1}{N}\sum_{u}\left|\widehat{T}_u - T_u\right|. \tag{A.3}$$

Thus, EM is higher-is-better and LenErr is lower-is-better.

KC normalized edit similarity (KC-NES).. Let $\mathrm{Lev}(\cdot,\cdot)$ be Levenshtein distance with unit insertion, deletion, and substitution costs. We compute

$$\mathrm{NES}_u = 1 - \frac{\mathrm{Lev}(\widehat{M}_u, M_u)}{\max(\widehat{T}_u, T_u, 1)}, \qquad \text{KC-NES} = \frac{1}{N}\sum_{u}\mathrm{NES}_u\,. \tag{A.4}$$

KC-NES lies in $[0,1]$; higher values indicate better joint agreement in state identity, order, and length. An invalid or missing trajectory receives zero under the fixed failure policy.

A.3 C–K Trajectory Personalization

KC-NES measures fidelity to the intended gold trajectory but does not by itself establish that the prediction is more specific to its owner than to a population-level trajectory. We therefore contrast each prediction with all gold trajectories in the cohort. Define

$$a_{uv} = 1 - \frac{\mathrm{Lev}(\widehat{M}_u, M_v)}{\max(\widehat{T}_u, T_v, 1)}. \tag{A.5}$$

KC own-user margin.

$$\Delta_u^{\mathrm{KC}} = a_{uu} - \frac{1}{N-1}\sum_{v\neq u} a_{uv}, \qquad \text{KC-Margin} = \frac{1}{N}\sum_{u}\Delta_u^{\mathrm{KC}}. \tag{A.6}$$

A positive margin indicates that the generated trajectory fits its intended participant better than an average non-target participant.

KC own-user AUC.. For each participant, we treat the own gold trajectory as the positive and each other participant's trajectory as a negative:

$$\mathrm{AUC}_u^{\mathrm{KC}} = \frac{1}{N-1}\sum_{v\neq u}\left(\mathbb{I}[a_{uu} > a_{uv}] + \frac{1}{2}\mathbb{I}[a_{uu} = a_{uv}]\right), \qquad \text{KC-AUC} = \frac{1}{N}\sum_{u}\mathrm{AUC}_u^{\mathrm{KC}}. \tag{A.7}$$

KC-AUC is bounded by $[0,1]$; 0.5 is the chance-ranking reference. KC-AUC and KC-Margin are the primary sequence-personalization metrics.

A.4 Trajectory Anti-Collapse and Geometry

Let

$$d(A,B) = 1 - \left[1 - \frac{\mathrm{Lev}(A,B)}{\max(|A|,|B|,1)}\right]$$

be normalized edit distance. Let $\overline{d}_{\mathrm{pred}}$ and $\overline{d}_{\mathrm{gold}}$ be its means over all unordered participant pairs for predicted and gold trajectories, respectively.

KC diversity retention (KCDR)..

$$\mathrm{KCDR} = \frac{\overline{d}_{\mathrm{pred}}}{\overline{d}_{\mathrm{gold}}}. \tag{A.8}$$

Higher KCDR indicates greater retention of the cross-participant variation present in the gold trajectories, with 1 denoting equal average dispersion. Values above 1 indicate over-dispersion and therefore are not automatically evidence of better personalization.

Trajectory geometry correlation (TGC)..

$$\mathrm{TGC} = \rho_{\mathrm{Spearman}}\left(\{d(\widehat{M}_u, \widehat{M}_v)\}_{u<v}, \{d(M_u, M_v)\}_{u<v}\right). \tag{A.9}$$

TGC lies in $[-1,1]$ and tests whether the relative geometry of between-participant trajectory distances is preserved; higher is better.

Unique trajectories and top share. For each distinct predicted trajectory A, let $n_A = \sum_u \mathbb{I}[\widehat{M}_u = A]$. We report

$$\mathrm{Unique} = |\{A : n_A > 0\}|, \qquad \mathrm{TopShare} = \frac{\max_A n_A}{N}. \tag{A.10}$$

More unique trajectories and a lower TopShare indicate less exact mode collapse. Chain diversity equals 1 − TopShare and is therefore omitted from the main table as redundant. KCDR, TGC, Unique, and TopShare are cohort-level diagnostics: none alone establishes that the variation is assigned to the correct participants.

A.5 NLI-Based V30 Value Expression

A frozen two-pole NLI probe maps each generated response and each human gold response to one stance for every V30 dimension: {AGREE, DISAGREE, NOT-EXPRESSED}. Accordingly, "Human" below denotes automatic agreement with value claims extracted from the human response, not a manual rating. For participant u, let G_u and H_u be the sets of non-neutral (dimension, polarity) claims extracted from generated and human responses.

Human expression precision, recall, and F1. We micro-aggregate directional claims over participants:

$$\mathrm{TP} = \sum_u |G_u \cap H_u|, \qquad \mathrm{FP} = \sum_u |G_u \setminus H_u|, \qquad \mathrm{FN} = \sum_u |H_u \setminus G_u|,$$

$$P_{\mathrm{human}} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R_{\mathrm{human}} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F1_{\mathrm{human}} = \frac{2P_{\mathrm{human}}R_{\mathrm{human}}}{P_{\mathrm{human}}+R_{\mathrm{human}}}. \tag{A.11}$$

All three are higher-is-better. Matching a dimension with the wrong polarity does not count as a true positive. Undefined denominators are reported as missing rather than assigned an artificial perfect score.

Profile-polarity precision and expression coverage. The six-point V30 score is binarized as DISAGREE for scores 1–3 and AGREE for scores 4–6. Let p_{uj} be this profile polarity and g_{uj} the generated claim polarity. For participants with $|G_u| > 0$,

$$\mathrm{PPP}_u = \frac{1}{|G_u|} \sum_{(j,g_{uj}) \in G_u} \mathbb{I}[g_{uj} = p_{uj}]. \tag{A.12}$$

Profile-polarity precision is the mean of PPP_u over participants who express at least one claim. Expression coverage includes all participants and is

$$\mathrm{Coverage} = \frac{1}{N} \sum_{u} \frac{|G_u|}{30}. \tag{A.13}$$

PPP measures directional consistency conditional on expression, whereas coverage exposes systems that attain precision by expressing very little.

Own-profile advantage. Let

$$\mathrm{Fit}(G_u, p_v) = \frac{1}{|G_u|} \sum_{(j,g_{uj}) \in G_u} \mathbb{I}[g_{uj} = p_{vj}]$$

for $|G_u| > 0$. The participant-specific contrast is

$$\mathrm{OPA}_u = \mathrm{Fit}(G_u, p_u) - \frac{1}{N-1} \sum_{v \neq u} \mathrm{Fit}(G_u, p_v), \qquad \mathrm{OPA} = \mathrm{mean}_{u:|G_u|>0}\, \mathrm{OPA}_u. \tag{A.14}$$

A positive OPA means that the expressed value polarities fit the target profile better than the profiles of other participants on average. As with PPP, participants without generated claims are excluded, so OPA must be interpreted jointly with coverage.

A.6 Statistical Reporting

The participant is the sole unit of inference. All system comparisons use a paired participant bootstrap with 10,000 resamples; each resample draws 162 participants with replacement, recomputes the statistic for both systems, and reports the 2.5th and 97.5th percentiles of the paired difference as its 95% confidence interval. Participant-decomposable scores are averaged within each resample, while micro-aggregated value-expression scores are recomputed from the resampled claim counts. For KCDR, TGC, Unique, and TopShare, the complete cohort statistic is recomputed after participant resampling. Although KCDR and TGC contain participant pairs, these pairs are dependent and are never treated as independent observations. We report valid sample sizes and expression coverage wherever undefined or missing outputs can change the effective cohort.

A.7 Q7 Human Alignment and Response Diversity

Human Reasoning Alignment. Two blinded raters assess how well each generated Q7 reasoning response matches the target participant's gold response. Using a five-point scale, they evaluate agreement in the prioritization of competing considerations, the logic of the trade-off process, and the overall decision direction. For participant u, the human-alignment score is

$$H_u = \frac{r_{u1} + r_{u2}}{2}, \tag{A.15}$$

where r_{u1} and r_{u2} are the ratings assigned by the two raters. We report the participant-level mean, a participant-bootstrap 95% confidence interval, and inter-rater reliability.

Response Uniqueness. Let R_u denote the generated Q7 reasoning response for participant u, and let $d_{uv} = \cos(e(R_u), e(R_v))$ denote the semantic similarity between the generated responses of participants u and v. We define between-participant response uniqueness as

$$\mathrm{RU} = 1 - \frac{2}{N_{\mathrm{Q7}}(N_{\mathrm{Q7}} - 1)} \sum_{\substack{u<v \\ u,v \in \mathcal{U}_{\mathrm{Q7}}}} d_{uv}, \tag{A.16}$$

where $\mathcal{U}_{\mathrm{Q7}}$ is the set of participants with valid generated Q7 reasoning responses and $N_{\mathrm{Q7}} = |\mathcal{U}_{\mathrm{Q7}}|$. Higher values indicate greater semantic variation across participants' generated responses. Response uniqueness is interpreted only as an anti-collapse diagnostic; it does not by itself establish alignment with the intended participants.

A.8 Statistical Reporting

The participant is the sole unit of inference. For system comparisons, we apply a participant-level paired bootstrap with 10,000 resamples and report percentile 95% confidence intervals for the paired difference. When multiple random seeds are used, their outcomes are first aggregated within participant. Human ratings additionally report quadratic-weighted Cohen's κ, ICC(2,1), and the effective number of participants. Value-semantic results always report coverage and effective sample size; a conditional mean alone is not used for cross-system conclusions.

B Detailed Trait-Conditioned Rule Induction and V30 Selection

This appendix gives the implementation-level methodology underlying Section 4. It describes how trait-conditioned rules are induced from Q1–Q6 and how V30 information is selected for a particular reasoning step. The six C–K modes and the Q1–Q7 protocol are defined in Section 3.3 and Section 3.4.2; Q7 is excluded from every procedure described below.

B.1 Shared Data Protocol

For participant i, let $\mathbf{v}_i = (v_{i1}, \ldots, v_{i30})$ denote the V30 profile, where each item is scored on a six-point scale. All regression models use the standardized profile $\widetilde{\mathbf{v}}_i$. Resampling, permutation tests, and cross-validation are performed at the participant level, so that all transitions from the same designer remain in the same split. We use 1,000 bootstrap samples, 1,000 participant permutations, and five participant-grouped folds. This protocol prevents repeated transitions from one designer from inflating statistical support.

The procedures produce exploratory routing candidates rather than confirmed psychological mechanisms. A candidate is retained only when its effect is stable under resampling and improves held-out prediction; the associated permutation and multiple-testing diagnostics determine its evidence level. At generation time, candidates are used as soft preferences, never as deterministic state transitions or causal claims.

Table 6: Trait-relation predicates used for pairwise rule induction.

Predicate	Activation condition
High co-activation	$v_{ij} \geq 4 \ \wedge \ v_{ik} \geq 4$
Low co-activation	$v_{ij} \leq 3 \ \wedge \ v_{ik} \leq 3$
Left tension	$v_{ij} \geq 4 \ \wedge \ v_{ik} \leq 3$
Right tension	$v_{ij} \leq 3 \ \wedge \ v_{ik} \geq 4$
Left dominance	$v_{ij} - v_{ik} \geq 2$
Right dominance	$v_{ik} - v_{ij} \geq 2$
Balance	$\lvert v_{ij} - v_{ik} \rvert \leq 1$

B.2 Single-Trait Transition Rules

For Q2, a trajectory $(z_{i1}, \ldots, z_{iT_i})$ is expanded with START and STOP boundaries. Repeated modes are retained. For each source mode s and target mode y, we fit a one-versus-rest L1-regularized logistic regression:

$$\operatorname{logit} P(z_{i,t+1} = y \mid z_{it} = s) = \alpha_{sy} + \boldsymbol{\beta}_{sy}^{\top} \widetilde{\mathbf{v}}_i + \gamma_{1,sy} \phi_{it} + \gamma_{2,sy} \phi_{it}^2, \quad \text{(B.1)}$$

where $\phi_{it} = 2t/T_i - 1$ is the centered position of a transition. The L1 penalty is selected separately for each source mode by participant-grouped cross-validation over $C \in \{0.03, 0.1, 0.3, 1.0, 3.0\}$.

For every (s, y, j) candidate, we record its standardized coefficient, participant support, bootstrap selection frequency, sign consistency, bootstrap interval, and held-out incremental log loss. A Q2 single-trait candidate enters the exploratory library when it has support from at least 20 participants, selection frequency of at least 0.90, sign consistency of at least 0.95, and $|\hat{\beta}_{sy,j}| \geq 0.15$. Held-out comparisons add one trait dimension to a position-only baseline; participant-level permutation tests and Benjamini–Hochberg correction quantify the remaining uncertainty.

Coefficient polarity is compiled into an interpretable profile condition. A positive coefficient yields a high-pole trigger ($v_{ij} \geq 4$); a negative coefficient yields a low-pole trigger ($v_{ij} \leq 3$). Both are rendered as a positive preference for the target operation under the corresponding profile condition. A negative coefficient therefore does not mean that the target operation is prohibited.

B.3 Trait-Relation Transition Rules

Single-trait effects cannot represent the possibility that two tendencies matter jointly. We first compute Spearman correlations for all unordered V30 pairs and retain stable pairs with $|\rho| \geq 0.20$, bootstrap sign consistency of at least 0.90, and a bootstrap interval that excludes zero. The 12 strongest retained pairs define the relation-candidate set.

For each pair (j, k), we construct the predicates in Table 6. A predicate is considered only when both the predicate and its complement contain at least 20 participants.

For a source–target pair and relation predicate R_{jk}, the baseline includes trajectory position and both trait main effects. The incremental model adds only the relation predicate:

$$\begin{aligned} M_0: \quad & \operatorname{logit} P(y = 1) = \alpha + \gamma_1 \phi + \gamma_2 \phi^2 + \beta_j \widetilde{v}_j + \beta_k \widetilde{v}_k, \\ M_1: \quad & M_1 = M_0 + \theta R_{jk}(\mathbf{v}_i). \end{aligned} \quad \text{(B.2)}$$

Thus, a retained relation rule reflects information beyond the two single-trait effects. We require positive residual support, bootstrap sign consistency of at least 0.90, improved held-out loss in at least three of five folds, and a favorable aggregate held-out loss. Negative relation effects are retained for analysis but are not compiled into generation preferences, because the routing language has no symmetric negative-action operator.

B.4 Initial and Contextual Preference Rules

Q1 records an initial mode rather than a transition. We therefore model $P(z_{i,\text{initial}} = y \mid \widetilde{\mathbf{v}}_i)$ and compile an initial-orientation preference, without creating a fictitious source mode. Q3–Q6 likewise provide one mode choice in a stated dilemma. For each question q and mode y, we fit

$$\operatorname{logit} P(Y_{iq} = y) = \alpha_{qy} + \boldsymbol{\beta}_{qy}^{\top} \widetilde{\mathbf{v}}_i. \quad \text{(B.3)}$$

The Q1 and Q3–Q6 models use a pre-specified L1 penalty of $C = 0.3$. Their exploratory screen requires at least 15 supporting participants, selection frequency of at least 0.80, sign consistency of at least 0.90, and $|\hat{\beta}| \geq 0.10$, together with favorable held-out diagnostics. Q1 produces initial-orientation candidates; Q3–Q6 candidates remain tied to their original conflict context and are activated only for semantically relevant situations.

B.5 Rule Compilation and Runtime Routing

Each retained candidate is compiled into a structured rule containing: (1) a V30 trigger, (2) a source operation or conflict context, (3) a preferred C–K operation, and (4) a factuality constraint. Statistical evidence is retained for audit but is not exposed to the language model. At generation time, routing is defined as

$$A_i = \text{Route}(V_i, \mathcal{R}, X^*), \quad \text{(B.4)}$$

where $\mathcal{R}$ is the fixed exploratory candidate set. The router first removes rules whose profile trigger or semantic context does not apply, then ranks the remaining rules by trigger strength and held-out improvement. It retains at most one rule from each applicable rule type. If a single-trait and a trait-relation rule conflict for the same source operation, the relation rule is preferred because it controls for the two single-trait effects.

Rules do not edit the generated sequence. They only state preferred operations, and public task facts override them. Two constraints are attached to every applicable rule: a $C \rightarrow K$ operation must be expressed as future or conditional verification, and a $K \rightarrow K$ operation may infer consequences from known information but may not introduce a new proposal.

B.6 Step-Conditioned V30 Cue Selection

The full V30 profile supplies stable individual context, but not every trait is equally relevant to every operation in a trajectory. We therefore use a second, step-conditioned channel during response realization. It is applied only after a C–K trajectory $M_i = (m_{i1}, \ldots, m_{iT})$ has been fixed. For step t, the selector returns eight V30 cues:

$$D_{it}^{(8)} = \text{Top8}_{d \in I^{Q2}} \, g\big(v_{id}, X^*, \text{pos}_t, \text{role}_t, m_{i,t-1}, m_{it}, m_{i,t+1}\big), \quad \text{(B.5)}$$

where I^{Q2} is a fixed set of Q2-relevant trait candidates. The selector evaluates each candidate against the public situation, the step's opening/process/boundary role, and the local transition $(m_{i,t-1}, m_{it}, m_{i,t+1})$. Public-situation relevance is ranked first; segment role and local

transition further disambiguate relevance; the designer's V30 direction and strength break remaining ties. The selected cues are rendered as directional value statements rather than item identifiers or numerical scores.

This mechanism answers a narrow question: which parts of a known profile are most relevant when realizing the current operation? It does not select the next C–K mode and cannot revise the fixed trajectory. Its purpose is to vary the concerns, trade-offs, and boundaries expressed at different steps while preserving the designer's complete profile as background context.

B.7 Exploratory Rule Library Summary

Table 7 summarizes the rule families used by the routing module. The table reports retained exploratory candidates rather than validated psychological laws. Full rule records, including anonymous triggers and stability diagnostics, are released as machine-readable supplementary material.

All retained candidates are used as soft operational priors. They may be overridden by public task facts and never directly edit a generated trajectory. The Q2 transition and Q3–Q6 contextual families form the 90-rule library used for the Q7 reasoning response; Q1 candidates are used only for initial-response settings. Rule discovery uses Q1–Q6 only; the held-out Q7 responses are excluded from rule mining and rule selection.

C R2 Conditional Generation Templates

This appendix documents the templates used by the R2 conditional generator. The production templates are administered in Chinese because the target responses are Chinese. The English templates below are a faithful, typesettable rendering of their instructions. They expose the shared instructions, message order, typed input blocks, and output contracts, while omitting participants' profiles, prior responses, and retrieved examples.

C.1 Two-Phase Message Flow

R2 uses two continuations in one conversation. Phase 1 produces only an observable six-mode C–K trajectory. After this trajectory is fixed, a deterministic selector derives step-conditioned V30 cues. Phase 2 then produces a Chinese response aligned with the fixed trajectory. No additional generative rewriting model is used.

Listing 1: Shared R2 instructions (English rendering).

```
You simulate an individual designer's observable
    reasoning and response
for research purposes. Work in two phases:
(1) produce only a six-mode C-K operation sequence
    ; then
(2) realize the fixed sequence as a first-person
    response.

The sequence describes observable operations in
    the response. It is not
hidden chain-of-thought and it is not an
    objectively optimal solution.

Priority of information:
1. The public design task is the sole source of
    current facts.
2. The target designer's professional context,
    value profile, dynamic
   value cues, and Q2 record indicate professional
    context, concerns,
   trade-offs, and cross-task tendencies. They do
    not supply Q7 facts.
3. Trait-conditioned rules are soft preferences.
4. Other designers' examples only calibrate
    response form and possible
   process organization.

Do not fabricate personal experience, project
    history, interviews, tests,
data, or completed evidence. Do not copy facts,
    positions, wording,
solutions, or state orders from another designer's
     example.

K: state provided facts, experience, data, or
    constraints; do not derive.
C: state a concept, objective, or proposal.
KtoC: derive a previously unstated concept or
    proposal from known input.
CtoK: specify future or conditional evidence
    gathering for an existing
      concept or proposal; do not claim that it
    was completed.
KtoK: infer consequences, limits, relations, or
    conflicts from known input;
      do not introduce a new proposal.
CtoC: refine, modify, combine, add to, remove from
    , or replace an
      identifiable existing concept or proposal.

Value statements do not receive a separate label.
    They are expressed through
one of the six operations when they advance that
    operation.
```

C.2 Phase 1: Trajectory Generation

The Phase 1 request is assembled in the following order: same-task demonstrations when the transductive condition is enabled; optional cross-designer Q2 examples; the target designer's professional background; the static V30 profile when enabled; the target designer's Q2 record; active trait-conditioned rules; the public Q7 task; and the output request. The target participant's own Q7 response and annotations are excluded from every input block.

Listing 2: Phase 1 template.

```
[SAME-TASK DEMONSTRATIONS, IF ENABLED]
Other designers' examples are provided only to
    calibrate output format and
task granularity. Do not treat them as facts about
     the target designer.
```

Table 7: Summary of the trait-conditioned exploratory rule library. A rule specifies a profile condition, an applicable source operation or conflict context, and a preferred C–K operation.

Rule family	Source data	Candidate relation	Runtime interpretation	Retained
Initial-orientation rules	Q1	(trait pole) $\Rightarrow z_{\text{initial}} = y$	Prefer an initial operation for an immediate response. Single-trait and trait-relation candidates are retained separately because Q1 has no preceding operation.	6 + 12 = 18
Single-trait transition rules	Q2	(trait pole, $z_t = s$) $\Rightarrow z_{t+1} = y$	When the current operation is s, a high or low pole of one trait favors operation y.	12
Trait-relation transition rules	Q2	$(R_{jk}(\mathbf{v}_i), z_t = s) \Rightarrow z_{t+1} = y$	When a pair of traits is jointly activated, in tension, dominant, or balanced, adjust the next-operation preference beyond the two single-trait effects.	49
Contextual preference rules	Q3–Q6	(conflict context,trait pole) $\Rightarrow y$	Favor operation y for a semantically matching design conflict. These rules are not treated as transitions because the source questions contain one operation choice only.	29

Table 8: R2 message flow. Bracketed symbols denote instance-specific data.

Message	Content
System	Shared instructions in Listing 1.
User, Phase 1	Context blocks and the state-sequence request in Listing 2.
Assistant	`{"state_sequence":[...]}`.
User, Phase 2	Fixed sequence, step-conditioned V30 cues, and the realization request in Listing 4.
Assistant	`{"nl_units":[...]}`.

```
[CROSS-DESIGNER Q2 EXAMPLES, IF ENABLED]
These records illustrate possible process
    organization in a different task.
Do not transfer their facts, proposals, positions,
     wording, or state order.

[TARGET PROFESSIONAL BACKGROUND]
[STATIC V30 PROFILE, IF ENABLED]
[TARGET Q2 RECORD]
[ACTIVE TRAIT-CONDITIONED RULES, IF ANY]
[PUBLIC Q7 TASK]

Return only a valid JSON object:
{"state_sequence":["..."]}

The sequence must be nonempty and may contain only
     K, C, KtoC, CtoK,
KtoK, and CtoC. Do not output V, START, STOP,
    explanations, or a natural-
language answer. Determine the order from the
    target context and current
task; do not copy a demonstration sequence, impose
     a fixed pattern, or add
empty steps merely to cover labels. CtoC is
    permitted only after C, KtoC,
or CtoC has introduced a concept earlier in the
    sequence.
```

C.3 Step-Conditioned V30 Cues

Let $\hat{M}_i = (\hat{m}_{i1}, \ldots, \hat{m}_{iT})$ be the Phase 1 output. For every step t, the deterministic selector uses the public task, the opening/process/boundary role of the step, and the local transition $(\hat{m}_{i,t-1}, \hat{m}_{it}, \hat{m}_{i,t+1})$ to select eight directional V30 cues. The selected cues influence only the concerns and trade-offs expressed while realizing that step; they do not modify $\hat{M}_i$.

Listing 3: Typed interface for step-conditioned V30 cues.

```
[STEP-CONDITIONED VALUE CUES]
These cues were selected after the C-K sequence
    was fixed. Each group affects
only the concerns and trade-offs of its
    corresponding step. Do not repeat the
cues as a list and do not change the fixed
    sequence.

Step 1/T: START -> [STATE_1] -> [STATE_2_OR_STOP]
- [DIRECTIONAL_VALUE_CUE_1]
...
- [DIRECTIONAL_VALUE_CUE_8]

Step t/T: [PREVIOUS_STATE] -> [CURRENT_STATE] -> [
    NEXT_STATE]
- [DIRECTIONAL_VALUE_CUE_1]
...
- [DIRECTIONAL_VALUE_CUE_8]
```

C.4 Phase 2: Response Realization

Phase 2 receives the fixed trajectory, one semantic contract per operation, the static V30 profile when enabled, and the step-conditioned

cues in Listing 3. It produces an array whose elements correspond to the trajectory order.

Listing 4: Phase 2 template.

```
[FIXED C-K SEQUENCE]
["STATE_1","STATE_2",...,"STATE_T"]

[OPERATION CONTRACTS]
1. STATE_1: [SEMANTIC CONSTRAINT FOR STATE_1]
...
T. STATE_T: [SEMANTIC CONSTRAINT FOR STATE_T]

[STATIC V30 PROFILE, IF ENABLED]
[STEP-CONDITIONED VALUE CUES]

The preceding sequence is fixed. Return only a
    valid JSON object:
{"nl_units":["..."]}

The array must have the same length as the fixed
    sequence. Element t realizes
only the t-th operation and preserves its order.
    Sentence form, length, and
degree of detail may vary naturally; do not use a
    fixed template. Concatenating
the units in order must form a 200--400 Chinese-
    character, first-person
response. Do not mention C-K labels, value scales,
     rules, examples, the model,
or the instruction. Do not fabricate completed
    research, tests, data, or
personal experience.
```

C.5 Output Handling

The final response is the ordered concatenation of `nl_units`. The system performs only deterministic checks: JSON parseability, the six-mode label set, equality between the number of units and the number of fixed operations, and exclusion of target Q7 records from their own input. Semantic failures, including fabricated evidence or an incorrect C–K operation, are retained for evaluation rather than repaired by a second model.

C.6 Participants' information

The 162 participants' main design disciplines were interaction design (30.2%), industrial design (22.2%), product design (17.9%), UX design (12.3%), and service design (9.9%). Most participants had 1–3 years (29.6%) or 3–5 years (25.9%) of professional experience. Over half had a design-major background (54.9%). In terms of current role, 48.1% were designers and 24.7% were students. The most common work context was student course projects (59.9%). For project roles, 50.6% were students, 27.8% were execution designers, and 21.6% were lead designers.

C.7 Dimensions of 30 traits

The 30 dimensional design traits cover attitudes toward technology, design process, creativity, risk, ethics, teamwork, and social responsibility. Each dimension is treated as a value axis. Together, these dimensions provide a compact profile of a participant's design-related preferences and priorities.



Figure 5: Information of the 162 participants